KEN N. HORIGOME

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Education

CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY Cebu City, Philippines
Bachelor of Science in Computer Science Expected Graduation: May 2028

- Honors: Presidential Scholar (4 semesters)

Technical Projects

SafeDismiss System — Sole Developer *Jul. – Aug. 2026*

- Architected and implemented a secure web-based student dismissal tracking system using **Spring Boot (Java)** and **vanilla JavaScript.**
- Engineered end-to-end data encryption using **AES-256-GCM** across the Spring Boot backend and **Web Crypto API** frontend to secure student Personally Identifiable Information (PII) during network transmission.
- Executed custom SQL queries via Spring JdbcTemplate to manage dynamic section assignments, fetcher records, and gate release logs.
- Built real-time gate monitoring and polling mechanisms to track student release statuses across multiple exit points.
- Integrated camera-based **QR code scanning** for contactless fetcher identification and developed dynamic **CSS** visual status alerts (warning, danger) for unconfirmed dismissals.

TIX.org (Capstone Project) — Team of 5 *Apr. – May 2026*

- Built a structured Data Access Layer using **JDBC**, **MySQL**, and DAO design patterns (DepartmentDAO, JobDAO) to enforce role-based access control.
- Engineered modular **JavaFX** desktop interfaces using FXML views and custom management modals for user roles, department assignments, and security credentials.
- Implemented desktop session persistence and state management using Java Object **Serialization**, ensuring reliable cleanup upon user logout.

Atendo (Capstone Project) — Team of 4 *Nov. – Dec. 2025*

- Implemented **AES encryption** utility modules to encrypt and decrypt sensitive student ID data stored within QR codes.
- Developed camera scanning integration with continuous frame retrieval and manual control triggers for reliable **ID verification.**
- Built core **OOP domain models** (Event, EventGroup) to manage hierarchical event structures and attendance groupings.
- Developed CSV data persistence modules to handle local batch reads/writes, simplifying event configuration and database seeding.

Additional Projects

- **Vantrex** — ***PHP, MySQL*** | Course Project, Information Management: Architected a role-based complaint ticketing web application to streamline cross-departmental workflows for complainant, respondent, and administrative roles.
- **StudyBu** — ***Kotlin, Android (MVP)*** | Course Project, Platform Development: Developed a native study productivity app featuring a Pomodoro timer, dynamic task list with calendar scheduling, secure user authentication, and a gamified virtual pet.
- **AgentVacuum & SearchAgent** — ***C#, .NET WinForms*** | Course Project, Intelligent Systems: Implemented reflex vacuum-cleaner and maze-solving pathfinding agents applying foundational AI agent design concepts.
- **PokéBooth** — ***JavaScript, WebRTC, HTML5 Canvas API, CSS*** | Custom Web App: Engineered a client-side photobooth app streaming live video to capture and composite user snapshots into customizable trading card layouts.
- **SariKiosk** — ***16-bit x86 Assembly (TASM)*** **|** Course Project, Computer Organization & Architecture: Engineered a kiosk simulator utilizing DOS interrupts (`INT 21h`), raw register manipulation, manual stack frame management, and custom ASCII-numeric data conversion.

Leadership & Activities

- **Project Lead:** TIX.org & Atendo Capstone Teams: Managed sprint timelines, software architecture, and team deliverables across two capstone projects. *A.Y. 2025 – 2026*
- **Freelance Multimedia Designer & Photographer:** Directed commercial product photography, advertisement layouts, and video post-production for local clients and campus publications. *2024 – Present*
- **Graphic Designer:** The Technologian Student Press — CIT-U *A.Y. 2025 – 2026*
- **Participant:** APRU 2026 Data, AI & Policy APAC Hackathon *2026*

Technical Skills

Languages: Java, C++, C#, Kotlin, PHP, JavaScript, SQL, Assembly (TASM), HTML/CSS
Frameworks & Libraries: Spring Boot, JavaFX, Node.js, Next.js, Django, Blazor, Tailwind CSS
Databases: MySQL, PostgreSQL, JDBC (JdbcTemplate)
Tools & Security Concepts: Git, GitHub, Maven, AES-256-GCM Encryption, Web Crypto API, RESTful APIs, WebRTC

Certifications

- CIT-U Java Object-Oriented Programming Certification, CodeChum — Score: 100% *May 2026*
- CIT-U C Programming I Certification, CodeChum — Score: 80% *Apr. 2025*
- Graphic Design Essentials, Canva Design School *Sept. 2025*